Theorem If $\lim_{x\to x_0} f(x)$ exists, then it is unique; that is, if

$$\lim_{x\to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x\to x_0} f(x) = L_2,$$

then $L_1 = L_2$.

Proof: Assume that $\lim_{x\to x_0} f(x)$ exists.

By way of contradiction suppose that the exist

$L_1, L_2 \in \mathbb{R}$ with $L_1 \neq L_2$ but such that

$\lim_{x\to x_0} f(x) = L_1$ and $\lim_{x\to x_0} f(x) = L_2$

Set $\varepsilon = |L_1 - L_2|/3$. By the definition of a limit

there exists $\delta_1, \delta_2 > 0$ such that if $x_1, x_2 \in D_f$ - the domain of f.

with $|x_1 - x_0| < \delta_1$ and $|x_2 - x_0| < \delta_2$, then

$|f(x_1) - L_1| < \varepsilon$ and $|f(x_2) - L_2| < \varepsilon$.

Then, for $x \in D_f$ with $|x - x_0| < \min\{\delta_1, \delta_2\}$ we

conclude

$$|L_1 - L_2| = |L_1 - f(x) + f(x) - L_2| \leq |L_1 - f(x)| + |f(x) - L_2|$$

($L_1 - f(x) \to 0$)

Triangle inequality

$$< \varepsilon + \varepsilon = 2\varepsilon = \frac{2}{3}|L_1 - L_2|$$

↯ ∎

Theorem If

$$\lim_{x\to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x\to x_0} g(x) = L_2,$$

then

$$\lim_{x\to x_0} (f+g)(x) = L_1 + L_2,$$
$$\lim_{x\to x_0} (f-g)(x) = L_1 - L_2,$$
$$\lim_{x\to x_0} (fg)(x) = L_1 L_2,$$

and, if $L_2 \neq 0$,

$$\lim_{x\to x_0} \left(\frac{f}{g}\right)(x) = \frac{L_1}{L_2}.$$

Note how we do not write

$\lim_{x\to x_0} f(x) = f(x_0)$ since (a) this might not be true

(b) x_0 might not belong to the domain of f.

Proof: We will prove the first statement and leave the remaining ones as an exercise. Alternatively, please see the lecture notes.

Proof that if $\lim_{x\to x_0} f(x) = L_1$ and $\lim_{x\to x_0} g(x) = L_2$

then

$$\lim_{x\to x_0} (f+g)(x) = L_1 + L_2$$

If $\lim_{x\to x_0} f(x) = L_1$ and $\lim_{x\to x_0} g(x) = L_2$ then given $\varepsilon > 0$

then exist $\delta_1, \delta_2 > 0$ such that if $x \in D_f$, $y \in D_g$

with $|x - x_0|$ and $|y - x_0|$

$|f(x) - L_1| < \varepsilon/2$ and $|g(y) - L_2| < \varepsilon/2$

then if $x \in D_f \cap D_g$ with $|x - x_0| < \min\{\delta_1, \delta_2\}$

$$|(f+g)(x) - (L_1 + L_2)|$$
$$= |f(x) + g(x) - L_1 - L_2|$$
$$= |f(x) - L_1 + g(x) - L_2|$$

Triangle inequality ↳
$$\leq |f(x) - L_1| + |g(x) - L_2|$$
$$< \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

∎

Hint: To show $\lim_{x \to x_0} (f \cdot g)(x) = L_1 L_2$

Note

$$|f(x)g(x) - L_1 L_2| = |f(x)(g(x) - L_2) + L_2(f(x) - L_1)|$$

and but $|f(x)| \leq L_1 + \varepsilon$ for x "close to x_0"

Hint: To show $\lim_{x \to x_0} \left(\frac{f}{g}\right)(x) = \frac{L_1}{L_2}$

Note

$$\left| \frac{f(x)}{g(x)} - \frac{L_1}{L_2} \right| = \left| \frac{L_2 f(x) - L_1 g(x)}{g(x) L_2} \right|,$$

$$|L_2 f(x) - L_1 g(x)| = |L_2(f(x) - L_1) - L_1(g(x) - L_2)|.$$

and but $|g(x)| \geq L_2 - \varepsilon$ for x "close to x_0".

Example Find the values of the following limits.

$$\lim_{x\to 2} \frac{9-x^2}{x+1} \quad \text{and} \quad \lim_{x\to 2} (9-x^2)(x+1)$$

$\lim_{x\to 2} \frac{9-x^2}{x+1} = \frac{9-4}{2+1} = \frac{5}{3}$ - What we expect the value of the limit to be.

We know $\lim_{x\to x_0} c = c$ and $\lim_{x\to x_0} x = x_0$ where c is a constant, i.e. a fixed real number

Therefore, we may conclude that

$$\lim_{x\to 2} 9-x^2 = \left(\lim_{x\to 2} 9\right) - \left(\lim_{x\to 2} x\right)\left(\lim_{x\to 2} x\right)$$

$$= 9 - 2\cdot 2 = 9-4 = 5$$

$$\lim_{x\to 2} x+1 = \lim_{x\to 2} x + \lim_{x\to 2} 1 = 2+1 = 3$$

and the function $x \mapsto x+1$ is defined on an open neighbourhood of the point 2.

Then Sin $\lim_{x\to 2} x+1 = 3 \neq 0$ we have

$$\lim_{x\to 2} \frac{9-x^2}{x+1} = \frac{\lim_{x\to 2} 9-x^2}{\lim_{x\to 2} x+1} = \frac{5}{3}$$

Moreover, we may conclude that

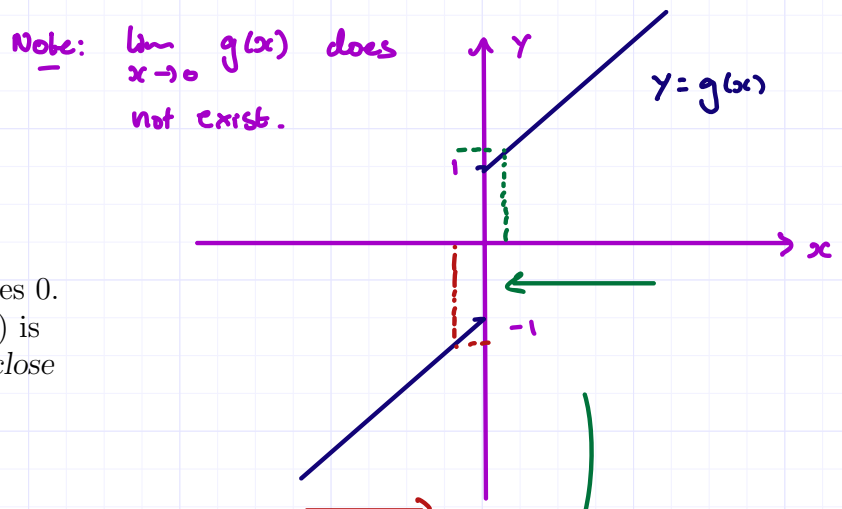
$$\lim_{x\to 2} (9-x^2)(x+1) = \left(\lim_{x\to 2} (9-x^2)\right)\left(\lim_{x\to 2} x+1\right) = 5\cdot 3 = 15.$$

One-sided limits

Consider the fucntion $g : \mathbb{R} \to \mathbb{R}$ defined by

$$g(x) = \begin{cases} x + 1 & \text{if } x \geq 0, \\ x - 1 & \text{if } x < 0. \end{cases}$$

Observe that $g(x)$ does not approach a limit as x approaches 0. since for arbitrarily small but positive x, we have that $f(x)$ is *close* to 1 but for arbitrarily small but negative x, $f(x)$ is *close* to -1. This motivates our next definition.



Definition

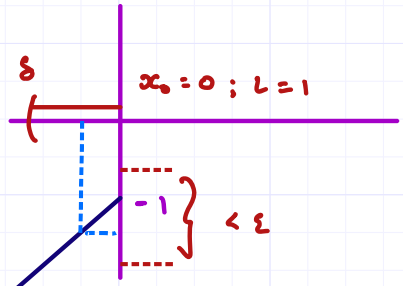
(a) We say that $f(x)$ **approaches the left-hand limit** L **as** x **approaches** x_0 **from the left**, and write

$$\lim_{x \to x_0 -} f(x) = L,$$

if f is defined on some open interval (a, x_0) and, for each $\epsilon > 0$, there is a $\delta > 0$ such that, if $x_0 - \delta < x < x_0$, then

$$|f(x) - L| < \epsilon$$

$x \in D_f$ - Domain of f.



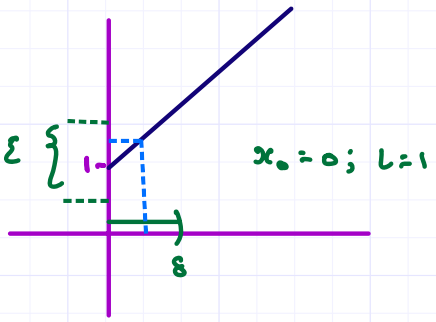
(b) We say that $f(x)$ **approaches the right-hand limit** L **as** x **approaches** x_0 **from the right**, and write

$$\lim_{x \to x_0 +} f(x) = L,$$

if f is defined on some open interval (x_0, b) and, for each $\epsilon > 0$, there is a $\delta > 0$ such that, if $x_0 < x < x_0 + \delta$

$$|f(x) - L| < \epsilon.$$

$x \in D_f$ - Domain of f.



Left- and right-hand limits are also called **one-sided limits**.

Example Compute $\lim_{x \to 0^-} g(x)$ and $\lim_{x \to 0^+} g(x)$

with $g: \mathbb{R} \to \mathbb{R}$

$$g(x) = \begin{cases} x+1 & \text{if } x \geq 0 \\ x-1 & \text{if } x < 0 \end{cases}$$

$\lim_{x \to 0^-} g(x)$ Let $\varepsilon > 0$ be given and let $\delta \in (0, \varepsilon)$. — To be filled in at the end of our calculation.

then for all $x \in (0-\delta, 0)$

Conj. $\lim_{x \to 0^-} g(x) = -1$ $\quad L = -1 \quad x_0 = 0$

$$|g(x) - (-1)| = |(x-1)\cdot(-1)|$$

$$= |x| < \delta < \varepsilon.$$ — To be filled in at the end of our calculation.

$\lim_{x \to 0^+} g(x)$ Let $\varepsilon > 0$ be given and let $\delta \in (0, \varepsilon)$

then for all $x \in (0, \delta)$

Conj. $\lim_{x \to 0^+} g(x) = 1$

$$|g(x) - 1| = |x + 1 - 1|$$

$$= |x| < \delta < \varepsilon$$

$L = 1 \quad x_0 = 0$

Theorem

A function f has a limit at x_0 if and only if it has left- and right-hand limits at x_0, and they are equal. More specifically,

$$\lim_{x \to x_0} f(x) = L$$

if and only if

$$\lim_{x \to x_0-} f(x) = \lim_{x \to x_0+} f(x).$$

The proof of this result is an application of the definition of a limit and a one sided limit.

Proof: "$\Rightarrow$" Suppose that $\lim_{x \to x_0} = L$, then

Given an $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \in D$ - the domain of f. with $|x - x_0| < \delta$,

$$|f(x) - L| < \varepsilon.$$

In particular, for all $x \in D$ with $x \in (x_0 - \delta, x_0)$

$$|f(x) - L| < \varepsilon$$

and for all $x \in D$ with $x \in (x_0, x_0 + \delta)$

$$|f(x) - L| < \varepsilon$$

Namely $\lim_{x \to x_0^-} f(x) = L = \lim_{x \to x_0^+} f(x)$

"$\Leftarrow$" Suppose that

$\lim_{x \to x_0^-} f(x) = \lim_{x \to x_0^+} f(x)$

and let us call this common value L. This means given an $\varepsilon > 0$ $\exists\ \delta_1, \delta_2$ such that for all $x \in (x_0 - \delta_1, x_0)$ and $y \in (x_0, x_0 + \delta_2)$

$|f(x) - L| < \varepsilon$ and $|f(y) - L| < \varepsilon$

Thus letting $\delta = \min\{\delta_1, \delta_2\}$ we have for all $z \in D$ with $z \neq x_0$ and $|z - x_0| < \delta$,

$$|f(z) - L| < \varepsilon$$

Hence,

$$\lim_{x \to x_0} f(x) = L.$$

$|z - x_0| < \delta$

Recall:

Theorem If $\lim_{x\to x_0} f(x)$ exists, then it is unique; that is, if

$$\lim_{x\to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x\to x_0} f(x) = L_2,$$

then $L_1 = L_2$.

Theorem

If

$$\lim_{x\to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x\to x_0} g(x) = L_2,$$

then

$$\lim_{x\to x_0} (f+g)(x) = L_1 + L_2,$$
$$\lim_{x\to x_0} (f-g)(x) = L_1 - L_2,$$
$$\lim_{x\to x_0} (fg)(x) = L_1 L_2,$$

and, if $L_2 \neq 0$,

$$\lim_{x\to x_0} \left(\frac{f}{g}\right)(x) = \frac{L_1}{L_2}.$$

Remark

With only minor modifications of their proofs (replacing the inequality $0 < |x - x_0| < \delta$ by $x_0 - \delta < x < x_0$ or $x_0 < x < x_0 + \delta$), it can be shown that the assertions of the above two theorems remain valid if "$\lim_{x\to x_0}$" is replaced by "$\lim_{x\to x_0-}$" (or "$\lim_{x\to x_0+}$") throughout.

$\lim_{x\to x_0^+} f(x) = L_1$ and $\lim_{x\to x_0^+} f(x) = L_2$

then $L_1 = L_2$

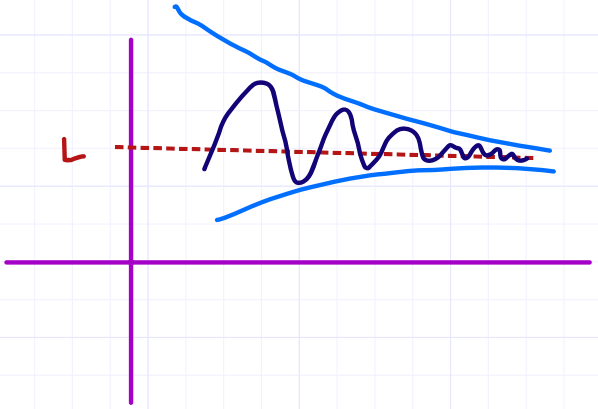
$\lim_{x\to x_0^+} f(x) = L_1$ and $\lim_{x\to x_0^+} g(x) = L_2$

then

$$\lim_{x\to x_0^+} f(x) + g(x) = L_1 + L_2$$

Limits at $\pm\infty$

Limits and one-sided limits have to do with the behaviour of a function f near to some point of D – the domain of f. It is equally reasonable to study f for large positive values of x if D is unbounded above or for large negative values of x if D_f is unbounded below.



Definition

We say $f(x)$ **approaches the limit** L **as** x **approaches** ∞, and write

$$\lim_{x\to\infty} f(x) = L,$$

if f is defined on an interval (a,∞) and, for each $\epsilon > 0$, there exists a real number β such that, for all $x > \beta$,

$$|f(x) - L| < \epsilon.$$

Similarly, $f(x)$ **approaches the limit** L **as** x **approaches** $-\infty$, written

$$\lim_{x\to-\infty} f(x) = L,$$

if f is defined on an interval $(-\infty,b)$ and, for each $\epsilon > 0$, there exists a real number α such that, for all $x < \alpha$,

$$|f(x) - L| < \epsilon.$$

Example Let $f:\mathbb{R}\to\mathbb{R}$ be defined by $f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 - 1/x^2 & \text{if } x \neq 0 \end{cases}$

$g:\mathbb{R}\to\mathbb{R}$ be defined by $g(x) = \sin(x)$

Determine if the following limits exist and if they exist find their values.

$\lim_{x\to\infty} f(x)$ and $\lim_{x\to\infty} g(x)$.

Let us first look at $\lim_{x \to \infty} g(x) = \lim_{x \to \infty} \sin(x)$ Note we are only concerned with large values of x

Observe that $\sin(2\pi n + \frac{\pi}{2}) = 1$ and $\sin(2\pi n) = 0$ for all $n \in \mathbb{N}$

This implies that $\lim_{x \to \infty} g(x)$ does not exist.

Let us now consider $\lim_{x \to \infty} f(x) = \lim_{x \to \infty} 1 - 1/x^2$ Note we are only concerned with large values of x

Conj. $\lim_{x \to \infty} f(x) = \lim_{x \to \infty} 1 - 1/x^2 = 1$.

Let $\varepsilon > 0$ be given and let $\beta > \varepsilon^{-1/2}$

To be filled in at the end of our calculation.

for all $x \in \mathbb{R}$ with $x > \beta$ we have

$$|f(x) - 1| = |(1 - 1/x^2) - 1| = |1/x^2| = \frac{1}{x^2} < \frac{1}{\beta^2} < \varepsilon$$

Therefore, $\lim_{x \to \infty} f(x)$ exists and equals 1.

Infinite limits

The functions

$$f(x) = \frac{1}{x}, \quad g(x) = \frac{1}{x^2}, \quad p(x) = \sin\frac{1}{x},$$

and

$$q(x) = \frac{1}{x^2}\sin\frac{1}{x}$$

do not have limits, or even one-sided limits, at $x_0 = 0$. They fail to have limits in different ways:

- $f(x)$ increases beyond bound as x approaches 0 from the right and decreases beyond bound as x approaches 0 from the left;
- $g(x)$ increases beyond bound as x approaches zero;
- $p(x)$ oscillates with ever-increasing frequency as x approaches zero;
- $q(x)$ oscillates with ever-increasing amplitude and frequency as x approaches 0.

Definition

We say $f(x)$ **approaches ∞ as x approaches x_0 from the left**, and write

$$\lim_{x \to x_0 -} f(x) = \infty,$$

if f is defined on an interval (a, x_0) and, for each real number M, there is a $\delta > 0$ such that for all $x_0 - \delta < x < x_0$

$$f(x) > M.$$

$x \in D_f$ - Domain of f.

Here the domain of each of these functions is $\mathbb{R} - \{0\}$ and the codomain can be taken to be $\mathbb{R}$.

There are 3 other kinds of infinite limits which are defined similarly.

- $\lim_{x \to x_0^+} f(x) = \infty$
- $\lim_{x \to x_0^-} f(x) = -\infty$
- $\lim_{x \to x_0^+} f(x) = -\infty$

If f is defined on an interval (x_0, b) and for each real number M there exists a $\delta > 0$ such that for all $x \in (x_0, b)$ with $x_0 < x < x_0 + \delta$ we have

$$f(x) < M.$$

Throughout, $\lim_{x\to x_0} f(x)$ **exists** will mean that

$$\lim_{x\to x_0} f(x) = L, \quad \text{where } L \text{ is } \textit{finite}.$$

To leave open the possibility that $L = \pm\infty$, we will say that $\lim_{x\to x_0} f(x)$ **exists in the extended reals**. This convention also applies to one-sided limits and limits as x tends to $\pm\infty$.

In this case we may write

$\lim_{x\mapsto x_0} f(x) = \infty$ or $\lim_{x\to x_0} f(x) = -\infty$.

Example

- $g : \mathbb{R} - \{0\} \to \mathbb{R}$ $\qquad g(x) = \frac{1}{x^2}$

Claim: $\lim_{x\mapsto 0^+} g(x) = \infty$.

Proof of claim: Let $M \in \mathbb{R}$ be given and let $\delta \in (0, 1/\sqrt{M})$

Observe that for $x \in (0,\infty)$ we $|x-0| < \delta$

we have

$$g(x) = \frac{1}{x^2} > \frac{1}{\delta^2} > M.$$

Exercise: Try computing the limits.

- $\lim_{x\mapsto 0^+} f(x)$
- $\lim_{x\mapsto 0^-} f(x)$

Observe that

$\lim_{x\mapsto 0^+} p(x)$ $\qquad$ $\lim_{x\mapsto 0^+} q(x)$

$\lim_{x\mapsto 0^-} p(x)$ $\qquad$ $\lim_{x\mapsto 0^-} q(x)$.

do not exist.

- Consider the function $h: \mathbb{R} \to \mathbb{R}$ defined by $h(x) = \sinh(x) = \dfrac{e^x - e^{-x}}{2}$

Compute the $\lim_{x \to \infty} \dfrac{e^x - e^{-x}}{2}$

Here we observe that $\lim_{x \to \infty} e^{-x} = 0$ and $\lim_{x \to \infty} e^x = \infty$ } Try showing this using our definitions above.

Thus we may conclude that

$$\lim_{x \to \infty} \frac{e^x - e^{-x}}{2} = \infty.$$

Here we have used properties of limits

Now consider the function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = e^{2x} - e^x$

Note: We can not obtain $\lim_{x \to \infty} e^{2x} - e^x$ by writing.

$$\lim_{x \to \infty} e^{2x} - e^x = \lim_{x \to \infty} e^{2x} - \lim_{x \to \infty} e^x$$

} Not allowed.

Since we obtain an indeterminate form.

These two limits do not exist.

However we can write

$$\lim_{x \to \infty} f(x) = \lim_{x \to \infty} e^{2x} - e^x = \lim_{x \to \infty} e^{2x}(1 - e^{-x})$$

Exercise: using our definitions above show that $\lim_{x \to \infty} e^{2x} = \infty$ and $\lim_{x \to \infty} 1 - e^{-x} = 1$

$$= \left(\lim_{x \to \infty} e^{2x}\right)\left(\lim (1 - e^{-x})\right) = \infty$$

$= \infty$ $\quad$ $= 1$

Monotonic functions

A function f is **non-decreasing** on an interval I if

$$f(x_1) \leq f(x_2),$$

whenever x_1 and x_2 are in I and $x_1 < x_2$. Similarly we say that f is **non-increasing** on I if

$$f(x_1) \geq f(x_2).$$

whenever x_1 and x_2 are in I and $x_1 < x_2$.

A function f is (**strictly**) **decreasing** on an interval I if

$$f(x_1) > f(x_2),$$

whenever x_1 and x_2 are in I and $x_1 < x_2$. Similarly, f is **increasing** on I if

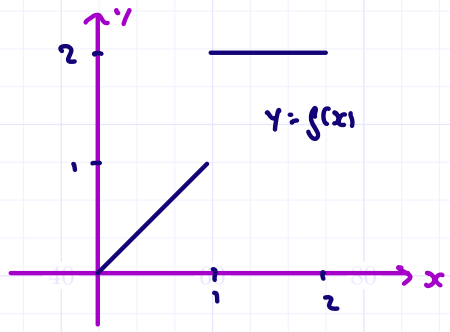
$$f(x_1) < f(x_2).$$

whenever x_1 and x_2 are in I and $x_1 < x_2$.

In any of these four cases, we call f **monotonic** on I.

Example

$f: [0,2] \to \mathbb{R}$ $\quad f(x) = \begin{cases} x & 0 \leq x < 1 \\ 1 & 1 \leq x \leq 2 \end{cases}$



this is a non-decreasing function.

if $x, y \in [0,1]$ $\quad x < y$

then $f(y) - f(x) = y - x > 0 \Rightarrow f(y) > f(x)$

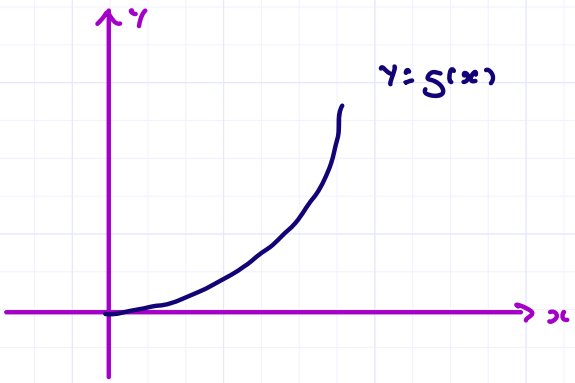
if $x, y \in [1,2]$ $\quad x < y$

th $f(y) - f(x) = 2 - 2 = 0 \Rightarrow f(y) \geq f(x)$

if $x \in [0,1]$ and $y \in [1,2]$

$\Rightarrow f(y) - f(x) = 2 - x > 0 \Rightarrow f(y) > f(x)$.

$g: [0,\infty) \to \mathbb{R}$ $\quad g(x) = x^2$



This is a strictly increasing function.

If $x, y \in [0,\infty)$ $\quad x < y$

then $f(y) - f(x) = y^2 - x^2 > 0$

$\Rightarrow f(y) > f(x)$.

Question: is $g: \mathbb{R} \to \mathbb{R}$ $\quad g(x) = x^2$ monotonic?

Exercise: Show that $h: \mathbb{R} \to \mathbb{R}$ $\quad h(x) = -x^3$ is strictly decreasing.

Theorem $f:(a,b)\to\mathbb{R}$

Suppose that f is monotonic on an interval (a,b) and define

$$\alpha = \inf_{a<x<b} f(x) \quad \text{and} \quad \beta = \sup_{a<x<b} f(x).$$

(a) If f is nondecreasing, then

$$\lim_{x\to a^+} f(x) = \alpha \quad \text{and} \quad \lim_{x\to b^-} f(x) = \beta$$

(b) If f is non-increasing, then

$$\lim_{x\to a^+} f(x) = \beta \quad \text{and} \quad \lim_{x\to b^-} f(x) = \alpha$$

(c) If $a < x_0 < b$, then

$$\lim_{x\to x_0^+} f(x) \quad \text{and} \quad \lim_{x\to x_0^-} f(x)$$

exist and are finite. Moreover,

$$\lim_{x\to x_0^-} f(x) \le f(x_0) \le \lim_{x\to x_0^+} f(x)$$

if f is non-decreasing, and

$$\lim_{x\to x_0^-} f(x) \ge f(x_0) \ge \lim_{x\to x_0^+} f(x)$$

if f is non-increasing.

Proof: We will prove, for f non-decreasing on (a,b),

- $\lim_{x\to a^+} f(x) = \alpha$ - the second part of (a) and part (b) follow from a similar argument
- If $a < x_0 < b$ then

$\lim_{x\to x_0^+} f(x)$ and $\lim_{x\to x_0^-} f(x)$

existence are finite, and that

$\lim_{x\to x_0^+} f(x) \le f(x_0) \le \lim_{x\to x_0^-} f(x)$

Proof that $\lim_{x\to a^+} f(x) = \alpha$.

Observe that if $M > \alpha$, there exists $x_0 \in (a,b)$

such that $f(x_0) < M$ - by definition of infimum.

Since f is non-decreasing this implies that

$f(x) < M$ if $a < x < x_0$.

Thus, if $\alpha = -\infty$, then

$\lim_{x\to a^+} f(x) = -\infty$.

Note that α cannot be ∞ as our codomain is $\mathbb{R}$.

Thus we now only need to consider the case

when $\alpha \in \mathbb{R}$.

To this end let $\varepsilon > 0$ be given and let

$M = \alpha + \varepsilon$. In which case $\forall x \in (a, x_0)$

$$\alpha \leq f(x) < \alpha + \varepsilon \iff |f(x) - \alpha| < \varepsilon.$$

i.e. $\lim_{x \to a^+} f(x) = \alpha$.

Prop. that if $a < x_0 < b$ then

$$\lim_{x \to x_0^+} f(x) \quad \text{and} \quad \lim_{x \to x_0^-} f(x)$$

exist and are finite, and that

$$\lim_{x \to x_0^+} f(x) \leq f(x_0) \leq \lim_{x \to x_0^-} f(x)$$

Apply part (a) to f on (a, x_0) and

(x_0, b) and using the codomain of f is $\mathbb{R}$

yields

$$\lim_{x \to x_0^+} f(x) = \inf_{x_0 < x < b} f(x) \in \mathbb{R}.$$

and

$$\lim_{x \to x_0^-} f(x) = \sup_{a < x < x_0} f(x) \in \mathbb{R}.$$

However, if $x_1, x_2 \in (a, b)$ are such that

$x_1 < x_0 < x_2$ then by the non-decreasing property

$$f(x_1) \leq f(x_0) \leq f(x_2)$$

Thus $\lim_{x \to x_0^-} f(x) \leq f(x_0) \leq \lim_{x \to x_0^+} f(x)$ ∎

2.2 Continuity

The definitions of

$$\lim_{x \to x_0} f(x), \quad \lim_{x \to x_0^-} f(x), \quad \text{and} \quad \lim_{x \to x_0^+} f(x),$$

do not involve $f(x_0)$ or even require that it be defined. However, the case where $f(x_0)$ is defined and equal to one or more of these quantities is important and relates to the concept of continuity.

Definition

(a) We say that f is **continuous at** x_0 if f is defined on an open interval (a, b) containing x_0 and

$$\lim_{x \to x_0} f(x) = f(x_0).$$

(b) We say that f is **continuous from the left at** x_0 if f is defined on an open interval (a, x_0) and

$$\lim_{x \to x_0-} f(x) = f(x_0).$$

(c) We say that f is **continuous from the right at** x_0 if f is defined on an open interval (x_0, b) and

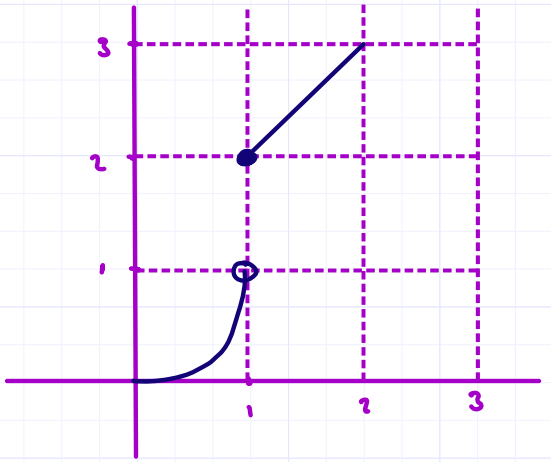
$$\lim_{x \to x_0+} f(x) = f(x_0).$$

Example

Determine at which points the function $f : [0, 2] \to \mathbb{R}$ defined, for $x \in [0,2]$, by

$$f(x) = \begin{cases} x^2 & \text{if } 0 \le x < 1, \\ x + 1 & \text{if } 1 \le x \le 2. \end{cases}$$

is continuous.



By inspection (one should prove this using our definition of limits) - left as an exercise.

$\lim_{x \to 0^+} f(x) = 0$

$\lim_{x \to 1^-} f(x) = 1$

$\lim_{x \to 1^+} f(x) = 2$

$\lim_{x \to 2^-} f(x) = 3$

Therefore, f is continuous from the left at 2

continuous from the right at 0, 1

Further, if $x_0, x \in (0,1)$ with $|x_0 - x| < \varepsilon/2$

$$|f(x) - f(x_0)| = |x^2 - x_0^2| = |x - x_0||x + x_0| \le 2|x - x_0| < 2 \cdot \varepsilon/2 = \varepsilon$$

$\Rightarrow \lim_{x \to x_0} f(x) = f(x_0)$ i.e. f is continuous at x_0.

If $x_0, x \in (1,2)$ with $|x_0 - x| < \varepsilon$

$$|f(x) - f(x_0)| = |(x+1) - (x_0+1)| = |x - x_0| < \varepsilon.$$

$\Rightarrow \lim_{x \to x_0} f(x) = f(x_0)$.

Definition

- A function f is **continuous on an open interval** (a, b) if it is continuous at every point in (a, b).
- If a function f is continuous on (a, b) and

$$\lim_{x \to b^-} f(x) = f(b),$$

then we say that f is **continuous on** $(a, b]$.

- If a function f is continuous on (a, b) and

$$\lim_{x \to a^+} f(x) = f(a),$$

then we say that f is **continuous on** $[a, b)$.

- If f is continuous on (a, b) and

$$\lim_{x \to b^-} f(x) = f(b) \quad \text{and} \quad \lim_{x \to a^+} f(x) = f(a),$$

then we say that f **is continuous on** $[a, b]$.

- More generally, if f is a function with domain D, and if S is a subset of D, consisting of finitely or infinitely many disjoint intervals, then f is **continuous on** S if f is continuous on every interval in S.

Example

- The function $f:[0,2] \to \mathbb{R}$ defined, for $x \in [0,2]$, by

$$f(x) = \begin{cases} x^2 & \text{if } 0 \le x < 1, \\ x+1 & \text{if } 1 \le x \le 2, \end{cases}$$

is continuous on $[0,1)$, it is also continuous on $[1,2]$

but it is not continuous on $[0,2]$.

- The function $f:[0,\infty) \to \mathbb{R}$ defined, for $x \in [0,\infty)$, by

$$f(x) = \sqrt{x}$$

is continuous on $[0,\infty)$.

Proof: Observe that given $\varepsilon > 0$, if $x \in (0, \varepsilon^2)$

$$|f(x) - f(0)| = |\sqrt{x} - \sqrt{0}| = \sqrt{x} < \sqrt{\varepsilon^2} = \varepsilon. \quad \Rightarrow \lim_{x \to 0^+} f(x) = f(x_0)$$

if $x_0 \in (0,\infty)$, letting $\delta = \varepsilon\sqrt{x_0}$ Note the dependence on x_0.

we have, for $x \in [0,\infty)$ with $|x - x_0| < \delta$,

$$|f(x) - f(x_0)| = |\sqrt{x} - \sqrt{x_0}| = |\sqrt{x} - \sqrt{x_0}| \cdot \frac{|\sqrt{x} + \sqrt{x_0}|}{|\sqrt{x} + \sqrt{x_0}|}$$

$$= \frac{|(\sqrt{x} - \sqrt{x_0})(\sqrt{x} + \sqrt{x_0})|}{\sqrt{x} + \sqrt{x_0}} \le \frac{|x - x_0|}{\sqrt{x_0}} < \varepsilon.$$

$$\Rightarrow \lim_{x \to x_0} f(x) = f(x_0).$$

Definition A function f is **piecewise continuous** on $[a, b]$ if

(a) $\lim_{x \to x_0^+} f(x)$ exists for all x_0 in $[a, b)$;

(b) $\lim_{x \to x_0^-} f(x)$ exists for all x_0 in $(a, b]$;

(c) $\lim_{x \to x_0^-} f(x) = \lim_{x \to x_0^+} f(x)$ for all but finitely many $x_0 \in (a, b)$.

All three need to be satisfied.

If (c) fails to hold at some x_0 in (a, b), f is said to have a **jump discontinuity at** x_0.

Also, f has a **jump discontinuity at** a if

$$\lim_{x \to a^+} f(x) \neq f(a)$$

and a **jump discontinuity at** b if

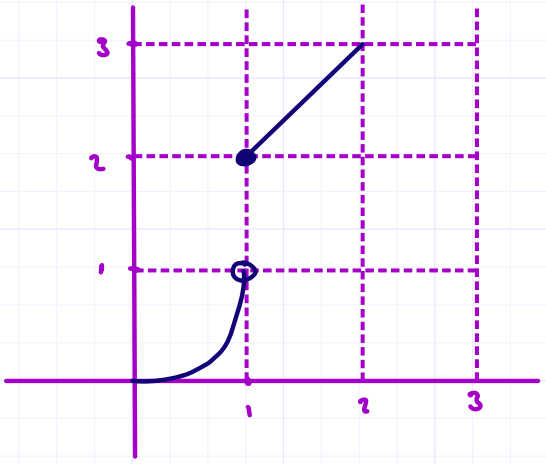
$$\lim_{x \to b^-} f(x) \neq f(b).$$

Example

- The function $f : [0,2] \to \mathbb{R}$ defined, for $x \in [0,2]$, by

$$f(x) = \begin{cases} x^2 & \text{if } 0 \leq x < 1, \\ x+1 & \text{if } 1 \leq x \leq 2, \end{cases}$$

is piecewise continuous with a jump discontinuity at 1.



Theorem If f and g are continuous on a set S, then so are $f+g$, $f-g$, and fg. In addition, f/g is continuous at each x_0 in S such that $g(x_0) \neq 0$.

Proof: We will prove that if f and g are continuous at a point x_0 then $f+g$ is continuous at x_0.

} With this at hand since f and g are continuous at all $x_0 \in S$, this yields that $f+g$ is continuous on S.

If f and g are continuous at x_0, then

$$f(x_0) = \lim_{x \to x_0} f(x) \quad \text{and} \quad g(x_0) = \lim_{x \to x_0} g(x)$$

$\Rightarrow$

$$(f+g)(x_0) = f(x_0) + g(x_0) = \lim_{x \to x_0} f(x) + \lim_{x \to x_0} g(x) = \lim_{x \to x_0} f(x) + g(x) = \lim_{x \to x_0} (f+g)(x)$$

By definition of $f+g$. By our above observation. Since both the limits exist. By definition of $f+g$. ∎

Remark With this result at hand, one may conclude that all polynomials are continuous and all well-defined rational functions are continuous.

Removable discontinuities

Let f be defined on a deleted neighborhood of x_0 and discontinuous (perhaps even undefined) at x_0. We say that f has a **removable discontinuity** at x_0 if $\lim_{x \to x_0} f(x)$ exists.

In this case, the function

$$g(x) = \begin{cases} f(x) & \text{if } x \in D \text{ and } x \neq x_0, \\ \lim_{x \to x_0} f(x) & \text{if } x = x_0, \end{cases}$$

is continuous at x_0.

Example

- The function $f : \mathbb{R} - \{0\} \to \mathbb{R}$ defined, for $x \in \mathbb{R} - \{0\}$, by

$$f(x) = x \sin(1/x)$$

is not defined at $x = 0$, and therefore not continuous at $x = 0$. However, we have seen that

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} x \sin(1/x) = 0$$

Hence, f has a removable discontinuity at $x = 0$.

- The function $g : \mathbb{R} - \{0\} \to \mathbb{R}$ defined, for $x \in \mathbb{R} - \{0\}$, by

$$g(x) = \sin(1/x)$$

is undefined at $x = 0$ and its discontinuity at $x = 0$ is not removable since $\lim_{x \to 0} g(x)$ does not exist in the reals, nor the extended reals.

Composite functions

We have seen that the investigation of limits and continuity can be simplified by regarding a given function as the result of addition, subtraction, multiplication, and division of simpler functions. Another operation useful in this connection is **composition** of functions; that is, the substitution of one function into another.

Definition

Suppose that f and g are functions with domains D_f and D_g. If D_g has a nonempty subset T such that $g(x) \in D_f$ whenever $x \in T$, then the **composite function** $f \circ g$ is defined on T by

$$(f \circ g)(x) = f(g(x)).$$

Example

If $f : (0, \infty) \to \mathbb{R}$ and $g : \mathbb{R} - \{-1, 1\} \to \mathbb{R}$ are defined by

$$f(x) = \log(x) \quad \text{and} \quad g(x) = \frac{1}{1-x^2}$$

Since $g(x) > 0 \iff x \in (-1, 1)$

the composite function $f \circ g$ has domain $(-1, 1)$

i.e. $f \circ g : (-1, 1) \to \mathbb{R}$ is defined by

$$f \circ g(x) = f(g(x)) = f\left(\frac{1}{1-x^2}\right) = \log\left(\frac{1}{1-x^2}\right)$$

Exercise: Find the domain of $g \circ g$.

Theorem Suppose that g is continuous at x_0, $g(x_0)$ is an interior point of D_f, and f is continuous at $g(x_0)$. Then $f \circ g$ is continuous at x_0.

Here D_f denotes the domain of f.

Proof: Aim: To show that

$$\lim_{x \to x_0} f \circ g(x) = f \circ g(x_0) = f(g(x_0))$$

↓ By definition.

Let $\varepsilon > 0$ be given.

Since $g(x_0)$ is an interior point of the domain of f, and f is continuous at $g(x_0)$, then exists a $\delta_1 > 0$ such that $(g(x_0) - \delta_1, g(x_0) + \delta_1) \subseteq D_f$ and for $t \in D_f$ with $|t - g(x_0)| < \delta_1$,

$$|f(t) - f(g(x_0))| < \varepsilon.$$

Since g is continuous at x_0, then exists a $\delta > 0$ such that for all $x \in D_g$ with $|x - x_0| < \delta$

$$|g(x) - g(x_0)| < \delta_1$$

D_g: the domain of g.

$\Rightarrow$ for all $x \in D_g$ with $|x - x_0| < \delta$

$$|f \circ g(x) - f \circ g(x_0)| = |f(g(x)) - f(g(x_0))| < \varepsilon$$

i.e. Let $t = g(x)$

■

Bounded functions

- A function f is **bounded below** on a set S if there exists a real number m such that, for all $x \in S$,

$$f(x) \geq m.$$

In this case, the set

$$V = \{f(x) \mid x \in S\}$$

has an infimum α, and we write

$$\alpha = \inf_{x \in S} f(x).$$

If there exists a point $x_1 \in S$ such that $f(x_1) = \alpha$, we say that α is the **minimum of f on S**, and write

$$\alpha = \min_{x \in S} f(x).$$

- Similarly, f is **bounded above on S** if there exists a real number M such that, for all x in S,

$$f(x) \leq M.$$

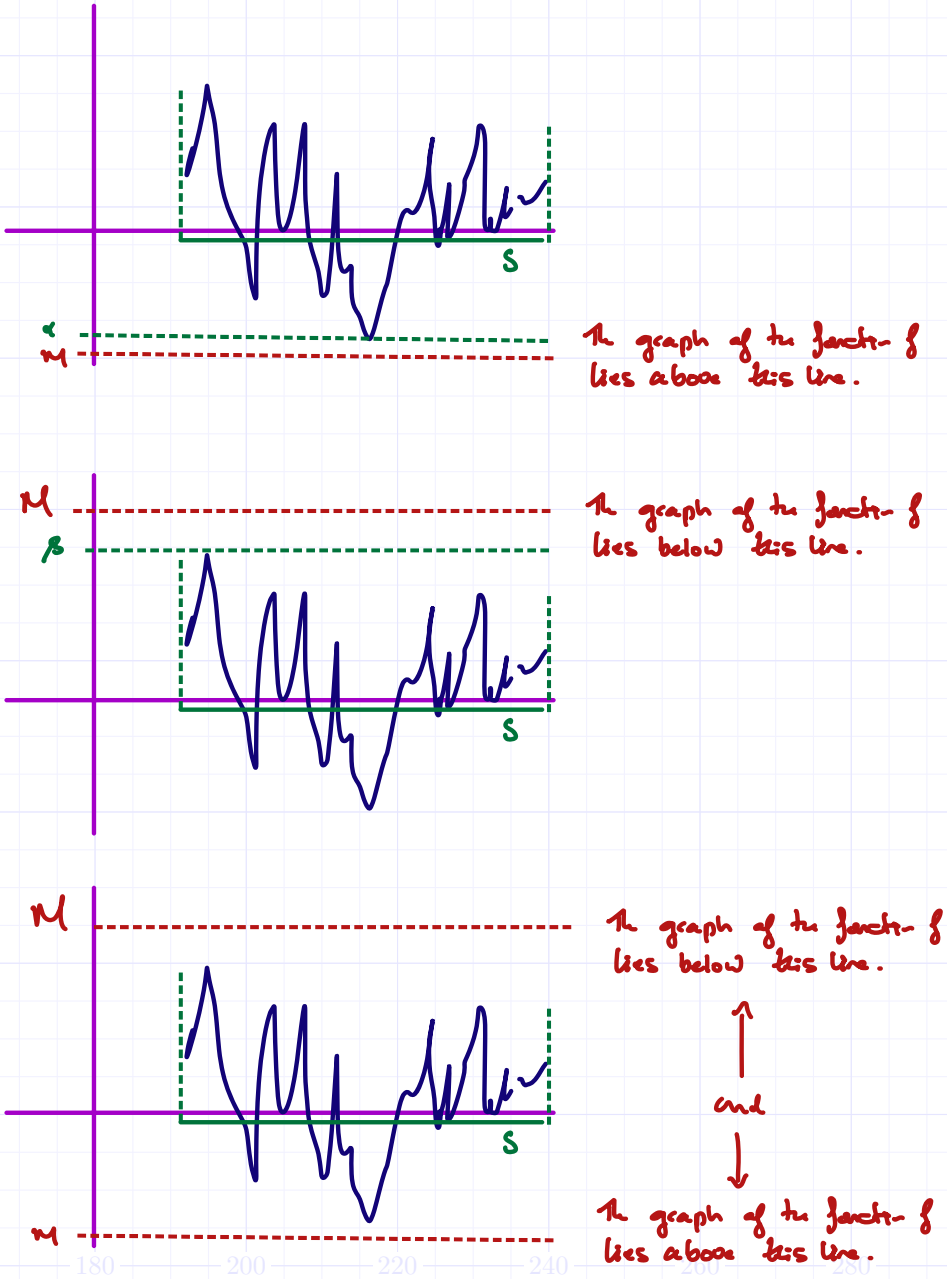
In this case, V has a supremum β, and we write

$$\beta = \sup_{x \in S} f(x).$$

If there exists $x_2 \in S$ such that $f(x_2) = \beta$, then we say that β is the **maximum of f on S**, and write

$$\beta = \max_{x \in S} f(x).$$

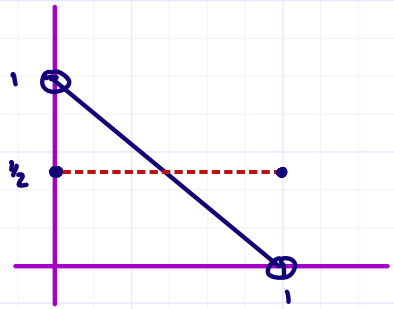
- If f is bounded above and below on a set S, we say that f is **bounded** on S.



Example

- The function $g: [0,1] \to \mathbb{R}$ defined, for $x \in [0,1]$, by

$$g(x) = \begin{cases} 1/2 & \text{if } x = 0 \text{ or } x = 1 \\ 1 - x & \text{if } 0 < x < 1 \end{cases}$$



is bounded on $[0,1]$, indeed

$$-1 \leq g(x) \leq 1/2 \quad \text{for all } x \in [0,1].$$

Moreover, we have that

$$\sup_{0 \leq x \leq 1} g(x) = 1 \quad \text{and} \quad \inf_{0 \leq x \leq 1} g(x) = 0$$

Thus, we may conclude that g has no minimum and no maximum.

There is a cute connection between continuity and boundedness which we discuss in our next result.

Theorem (real valued)

If f is continuous on a finite closed interval $[a,b]$, then f is bounded on $[a,b]$.

To prove this result we will use the idea of compactness - i.e. the Heine-Borel Thm.

Proof: Suppose that $t \in [a,b]$.

Since f is continous at t, there exists an open interval I_t containing t such that for all $x \in I_t \cap [a,b]$,

$$|f(x) - f(t)| < 1$$

Using the fact that $\lim_{x \to t} f(x) = f(t)$ $\quad t \in (a,b)$ and the ε-δ definition of limits with $\varepsilon = 1$.

The collection $\mathcal{U} = \{ I_t : t \in [a,b] \}$ is an open cover of $[a,b]$.

Since $[a,b]$ is compact, the Heine-Borel theorem implies there exists finitely many points $t_1, t_2 \ldots t_n$ such that

$$\bigcup_{i=1}^{n} I_{t_i} \supseteq [a,b]$$

Therefore, for $x \in [a,b]$ $\exists$ $i \in \{1,2 \ldots n\}$ with $x \in I_{t_i}$ and so

$$\begin{aligned} |f(x)| = |f(x) - f(t_i) + f(t_i)| &\leq |f(x) - f(t_i)| + |f(t_i)| \\ &\leq 1 + |f(t_i)| \\ &\leq 1 + \max_{i \in \{1,2 \ldots n\}} |f(t_i)| \end{aligned}$$

That is, setting $M = 1 + \max_{i \in \{1,2 \ldots n\}} |f(t_i)|$ we have, for all $x \in [a,b]$,

$$-M \leq f(x) \leq M.$$

∎

Theorem

Suppose that f is continuous on a finite closed interval $[a, b]$. If

$$\alpha = \inf_{a \le x \le b} f(x) \quad \text{and} \quad \beta = \sup_{a \le x \le b} f(x),$$

then α and β are respectively the minimum and maximum of f on $[a, b]$.

Namely, there exist $x_1, x_2 \in [a, b]$ with

$$f(x_1) = \alpha \quad \text{and} \quad f(x_2) = \beta.$$

Again, to prove this statement we will use compactness.

Proof: We will show that there exists $x_1 \in [a,b]$ such that $f(x_1) = \alpha$.

Suppose, by way of contradiction, that there exists no such x_1. In which case, $f(x) > \alpha$ for all $x \in [a,b]$.

Suppose that $t \in [a,b]$. Then $f(t) > \alpha$, and so

$$f(t) > \frac{f(t)+\alpha}{2} > \alpha$$

Since f is continuous at t there is an open interval I_t containing t such that for all $x \in I_t \cap [a,b]$

$$f(x) > \frac{f(t)+\alpha}{2}.$$

The collection $\mathcal{H} = \{ I_t \mid t \in [a,b] \}$ is an open cover of $[a,b]$.

Since $[a,b]$ is compact there exist $t_1, t_2 \dots t_n$ such that

$$\bigcup_{i=1}^{n} I_{t_i} \supseteq [a,b]$$

Set

$$\alpha_1 = \min_{i \in \{1,2,\dots n\}} \frac{f(t_i)+\alpha}{2}$$

This implies that $\forall\ t \in [a,b]$

$$f(t) > \alpha_1 > \alpha \quad \Rightarrow \quad \lightning$$

■